

The impact of Intellectual Property Rights Protection on Digital Industry Innovation

Zhenghao Chen*

School of Economics, Wuhan University of Technology,
Wuhan, China
czh9928@163.com*

Zuhai Tian

Trade Frictions and Barriers Warning Center of Hubei,
Wuhan, China
270200@whut.edu.cn

ABSTRACT

This paper empirically analyzes the impact of intellectual property rights protection on digital industry innovation through China's IPR model city policy as a quasi-natural experiment, based on panel data of 286 prefecture-level cities in China from 2005-2019, and concludes that the improvement of IPR protection significantly promotes digital industry innovation. Therefore, while developing digital industries, countries should improve their own intellectual property protection levels to better stimulate digital industries to innovate and improve digital technologies.

CCS CONCEPTS

• **Social and professional topics** → Computing / technology policy; Intellectual property.

KEYWORDS

Digital industry, innovation, intellectual property rights protection, Differences-in-Differences

ACM Reference Format:

Zhenghao Chen* and Zuhai Tian. 2023. The impact of Intellectual Property Rights Protection on Digital Industry Innovation. In *2023 4th International Conference on Big Data Economy and Information Management (BDEIM 2023)*, December 08–10, 2023, Zhengzhou, China. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3659211.3659300>

1 INTRODUCTION

Benefiting from the accelerated evolution of underlying digital technologies such as artificial intelligence, blockchain, cloud computing and big data, the digital economy is continuously integrated into all aspects of the whole process of economic development. As the core driver of high-quality development of digital economy, digital industry innovation has become a universal technology profoundly involved in economic and social development [1], is rapidly promoting structural changes in production, distribution, exchange and consumption [2], and playing a crucial role in whether the country can seize the first opportunity of digital industry development, create new advantages in digital economy and build new national competitive advantages.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

BDEIM 2023, December 08–10, 2023, Zhengzhou, China

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1666-9/23/12

<https://doi.org/10.1145/3659211.3659300>

However, the knowledge-intensive nature and rapid evolution of digital industry innovation can present a range of obstacles. These include challenges in verifying and tracking infringement, as well as the considerable difficulty in upholding rights [3]. This leads to a deficit of internal capacity for autonomous innovation within the digital industry sector, necessitating the implementation of a robust system for safeguarding intellectual property rights (IPR) that allows for the sustenance and enhancement of innovation [4].

Amidst the current trend of many countries actively expanding the digital industry and developing digital economies, this study examines the extent to which IPR protection can drive innovation within the digital industry sector. Specifically, we analyze China's model cities of intellectual property construction as a quasi-natural experiment, in order to explore this phenomenon's practical significance and research value.

2 LITERATURE REVIEW

The literature related to the study of this paper is divided into two main areas. The first is a study on digital industry innovation. Early research on innovation in the digital industry focused on the communications technology industry. As the importance of the digital economy becomes more and more pronounced, some scholars have explored the factors influencing digital industry innovation based on different perspectives, and believe that digital economic development, digital investment, and government policies are important influencing factors [5-7]. However, there is a lack of research on innovation in the digital industry, and the relationship between IPR protection and innovation in the digital industry, as an important tool for government to stimulate innovation, has not yet been explored by scholars.

The second is the research on the impact of IPR protection on innovation. A multitude of scholars have studied the impact of IPR protection. One perspective argues that an improved property rights system can not only decrease the frequency of technology imitation, safeguard the benefits derived from technological innovation, and stimulate intrinsic motivation for innovation [8] [9], but also enhance access to public patent information, reduce the cost of knowledge acquisition, and strengthen innovative capabilities [10]. Another perspective maintains that a reasonable range of IPR protection intensity is necessary for incentivizing technological innovation, and that excessive protection not only results in inefficient and redundant innovation efforts, but also reinforces technological monopolies and curtails innovation autonomy [11].

In summary, among the existing studies, there are fewer studies on digital industry innovation, and there is a shortage of studies exploring the impact of IPR protection on it; there are rich studies on the impact of IPR protection on innovation, but there are no

unified conclusions yet, and there is a lack of research perspectives on digital industries.

3 THEORY ANALYSIS

In the context of the digital age, digital industries are keen to pursue technological innovation to capitalize on opportunities, while also being mindful of potential risks associated with innovation. Institutional theory suggests that the intellectual property system can influence firms' incentives to sustain innovation by regulating innovation activities and reducing associated risks. Establishing and improving the intellectual property protection system can create a competitive market environment for innovation output, reduce the risk of infringement of innovation output in the digital industry, and provide a strong incentive for firms to sustain innovation. Additionally, promoting intellectual property protection can attract human and capital resources, thereby providing the digital industry with the motivation and resources to improve the level of technological innovation.

From an external perspective, the digital industry, as the primary source of innovation in high-tech industries, gains valuable technological insight from publicly available patent information. This avoids redundant innovation inputs at an early stage, aiding businesses in adapting their innovation strategies promptly. Intellectual property protection plays a significant role in preventing infringement behaviors, such as copying and imitation, by safeguarding the rights and interests of original innovators and incentivizing further innovation. A sound environment for intellectual property protection can attract high-quality foreign investment while promoting cooperation and exchange, resulting in improved innovation efficiency. Within the internal environment, intellectual property as an intangible property right, fueled by patenting technologies, patent industrialization, and product branding, yields substantial profits for digital industries to further reinvest in innovation, motivating enterprises to invest more in innovation. Enterprises have the desire and willingness to invest additional funds into innovative research and development activities. This will enhance the independent innovation capability of the digital industry, thus creating a positive cycle within the industry and cultivating competitive advantages. Furthermore, the allocation of innovation factors is impacted by the level of protection granted to intellectual property rights. A high-quality environment for property rights protection will promote superior human capital innovation input and enhance its mobility, elevating the efficiency of innovation factor allocation, and ultimately generating significant impetus for digital industry innovation.

Based on the above analysis, the hypothesis is as follows:

H1: The improvement of IPR protection can promote innovation in the digital industry.

4 DATA SOURCE AND SAMPLE SELECTION

4.1 Research design

In this paper, we employ the multi-stage Difference-Difference method to examine the effect of intellectual property rights (IPR) protection on innovation within the digital industry. To do so, we utilise the IPR model city policy as a quasi-natural experiment,

which serves as a representation of the enhancement of IPR protection. Firstly, this study identifies the city treatment group by determining whether the city has joined the "intellectual property protection model city". If the city is a part of the model, it becomes the treatment group, otherwise, it is considered as the control group. The research model is as follows:

$$Dig_{it} = \alpha_0 + \alpha_1 DID_{it} + \alpha_2 Control_{it} + \lambda_i + \lambda_t + \varepsilon_{it} \quad (1)$$

Where Dig_{it} denotes the level of digital industry innovation of city i at time t , DID_{it} denotes the enhancement of IPR protection, $Control_{it}$ reflects a set of control variables at the city level that may affect digital industry innovation, λ_i denotes the city fixed effect of city i that does not change over time, λ_t denotes the time fixed effect, ε_{it} is a random disturbance term, and α_1 measures the net effect of demonstration city construction on urban digital industry innovation.

4.2 Data description and selection of variables

Due to data availability and the impact of COVID-19 in 2020, this paper takes 286 prefecture-level cities in China from 2005 to 2019 as the study sample, with 57 IPR demonstration cities as the treatment group and 229 other IPR non-demonstration cities as the control group. The data of IPR demonstration cities are manually compiled, and the data of the explained variable are obtained from the patent data of the China Intellectual Property Office, while other data are mainly obtained from the China City Statistical Yearbook and EPS database. The specific selection of variables is described as follows:

Explained variable: digital industry innovation (Dig)

The measurement of digital industry innovation in each region is based on the number of patent applications for digital core industries per 10,000 people. The definition of digital core industries is taken from the "Statistical Classification of the Digital Economy and Its Core Industries (2021)" published by the National Bureau of Statistics of China.

Core explanatory variables: IPR protection (DID)

In this paper, the IPR model city policy variable is used as a proxy variable for the enhancement of IPR protection. Specifically, DID is the product of the dummy variables $Treat_i$ and $Post_t$, where $Treat_{it}$ is a treatment group identifying variable based on whether the city is a model city for IPR protection, which is 1 if it is enrolled, and 0 otherwise; and $Post_{it}$ is a temporal identifying variable, which is 0 before the implementation of the model city policy for IPR protection, and 1 after the implementation. As of 2019, a total of 77 cities have been approved as IPR demonstration cities. For this study, policy dummy variables are used as a measurement. To ensure accuracy in determining the experimental group, the prefecture-level cities where some IPR demonstration city pilots are county-level cities or only parts of cities were excluded from the analysis.

Control variables

This paper also controls for the following variables: The level of economic development ($\ln gdp$), expressed as the logarithmic value of regional GDP; Science input ($\ln SE$), expressed in terms of the logarithm of the amount of urban science spending; Infrastructure level ($\ln PR$), expressed in terms of roads per capita in the city; The level of Internet development ($\ln INT$), expressed as the logarithm of the number of international Internet users in the city; The level of per capita income ($\ln wage$), expressed as the logarithm of the

Table 1: Descriptive statistics

Variable	N	Mean	SD	Min	Max
Dig	2359	2.126	5.027	0	75.71
DID	2359	0.123	0.328	0	1
lngdp	2359	16.50	0.878	13.86	19.41
lnSE	2359	5.062	1.163	1.807	10.10
lnPR	2359	3.104	0.558	0.562	5.378
lnINT	2359	13.91	0.924	11.01	16.57
lnwage	2359	10.77	0.335	9.534	11.77

Table 2: Regression result

	(1)	(2)
	Dig	Dig
DID	3.896*** (0.452)	3.470*** (0.429)
IPP		
lngdp		3.021*** (0.465)
lnSE		0.689*** (0.108)
lnPR		-0.582** (0.253)
lnINT		-1.126*** (0.241)
lnwage		-2.024*** (0.576)
Constant	1.647*** (0.0694)	-12.39* (7.197)
City FE	YES	YES
Year FE	YES	YES
Observations	2,359	2,359
R-squared	0.824	0.835

average urban wage amount. The results of descriptive statistics for each variable are presented in Table 1.

improvement of IPR protection promotes innovation in the digital industry.

5 EMPIRICAL RESULTS

5.1 Baseline regression results

The results of the study are presented in Column (1) and Column (2) of 2. Regression result, which shows the estimated impact of IPR protection on digital industry innovation. Column (1) reports the results without controlling for other variables, while Column (2) includes control variables. The estimated coefficient of the core explanatory variable is 3.896 in Column (1), which is significantly positive at the 1% level, indicating that the improvement of IPR protection positively affects innovation in the digital industry. The addition of control variables in Column (2) shows some changes in the estimated coefficients, but the sign remains positive and significant at the 1% level, which supports hypothesis 1 that the

5.2 Robustness tests

1. Replacement of the core explanatory variables

In order to ensure the robustness of the findings, this paper takes the number of IPR trials concluded per 10,000 people (IPP) in cities as a proxy variable for IPR protection. The regression results are shown in column (1) of 3. IPR protection significantly promotes digital industry innovation, which verifies the robustness of the conclusions of this paper.

2. Instrumental variable method (IVM)

In order to further reduce the impact of endogeneity on the empirical conclusion, this paper selects the one-stage lag of the core variable as the instrumental variable, which not only ensures the correlation between the instrumental variable and the endogenous explanatory variable, but also alleviates the correlation between the

Table 3: Robustness tests result

	(1)	(2)
	Dig	Dig
DID		5.6100*** (9.3115)
IPP	2.016*** (0.270)	
lngdp	2.278*** (0.425)	2.7168*** (5.4096)
lnSE	0.563*** (0.105)	0.6660*** (6.0376)
lnPR	-0.404** (0.189)	-0.5171** (-2.0463)
lnINT	-0.864*** (0.192)	-0.9211*** (-3.4553)
lnwage	-1.667*** (0.517)	-1.7952*** (-2.7674)
Constant	2.016*** (0.270)	2.7168*** (5.4096)
City FE	YES	YES
Year FE	YES	YES
Kleibergen-Paap rk LM statistic		201.214 [0.0000]
Kleibergen-Paap rk Wald F statistic		1972.486 {16.38}
Observations	2,359	2327
R-squared	0.617	0.1332

core explanatory variable and the perturbation term. The regression results are shown in column (2) of 3. After the instrumental variable passes the unidentifiability test and the weak instrumental variable test, IPR protection significantly promotes digital industry innovation, which verifies the robustness of the conclusions of this paper.

3. Parallel trend test

As illustrated in 1, the findings indicate that there were no discernible divergences in the trends of digital industry innovation levels between the treatment and control groups of cities prior to the policy's implementation. However, a significant disparity emerged in the aftermath of the policy's implementation, thereby affirming the satisfaction of the parallel trend test.

4. Placebo test

In order to verify that the increase in the level of innovation in the digital industry in cities is brought about by the improvement in IPR protection rather than other random factors, the number of cities with the same number of IPR demonstration cities is randomly selected as the treatment group and the remaining cities as the control group based on model (1) for regression estimation. The results are shown in 2. The estimated coefficients of the model city policy variables are all around 0 and normally distributed, indicating that the impact of model city policy on digital autonomy is not driven by other random factors, further demonstrating that the baseline regression results are robust.

6 CONCLUSIONS

This study investigates the impact of protecting Intellectual Property Rights (IPR) on innovation in the digital sector, using China's IPR model pilot city policy as a quasi-natural experiment. Data analysis for 286 prefecture-level cities in China from 2005 to 2019 was conducted. The outcomes indicate that improved IPR protection has a substantial positive effect on innovation in the digital industry. To encourage innovation and advance digital technology, countries must enhance their intellectual property protection and integrate it into digital economy development. This will stimulate the intrinsic impetus for innovation in digital industries.

REFERENCES

- [1] Gopal, R. D., Ramesh, R., & Whinston, A. B. 2003. Microproducts in a digital economy: Trading small, gaining large. *International Journal of Electronic Commerce*, 8(2), 9-30. <https://doi.org/10.1080/10864415.2003.11044292>
- [2] Perez, C. 1983. Structural change and assimilation of new technologies in the economic and social systems. *Futures*, 15(5), 357-375. [https://doi.org/10.1016/0016-3287\(83\)90050-2](https://doi.org/10.1016/0016-3287(83)90050-2)
- [3] Ciocoiu, Carmen. 2011. Considerations about Intellectual Property Rights, Innovation and Economic Growth in the Digital Economy. *Economia. Seria Management*. 14. 310-323. https://www.researchgate.net/publication/227490116_Considerations_about_Intellectual_Property_Rights_Innovation_and_Economic_Growth_in_the_Digital_Economy
- [4] Gambardella, A., Heaton, S., Novelli, E., & Teece, D. J. 2021. Profiting from enabling technologies? *Strategy Science*, 6(1), 75-90. <https://doi.org/10.1287/stsc.2020.0119> https://www.researchgate.net/publication/227490116_Considerations_about_Intellectual_Property_Rights_Innovation_and_Economic_Growth_in_the_Digital_Economy
- [5] Liu, Li, Tao Ding, and Hao Wang. 2022. "Digital Economy, Technological Innovation and Green High-Quality Development of Industry: A Study Case of China"

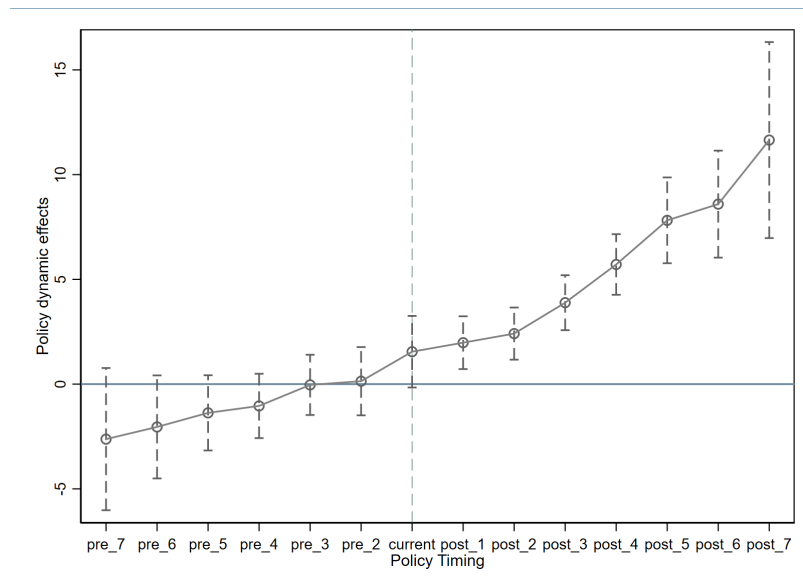


Figure 1: Parallel trend test

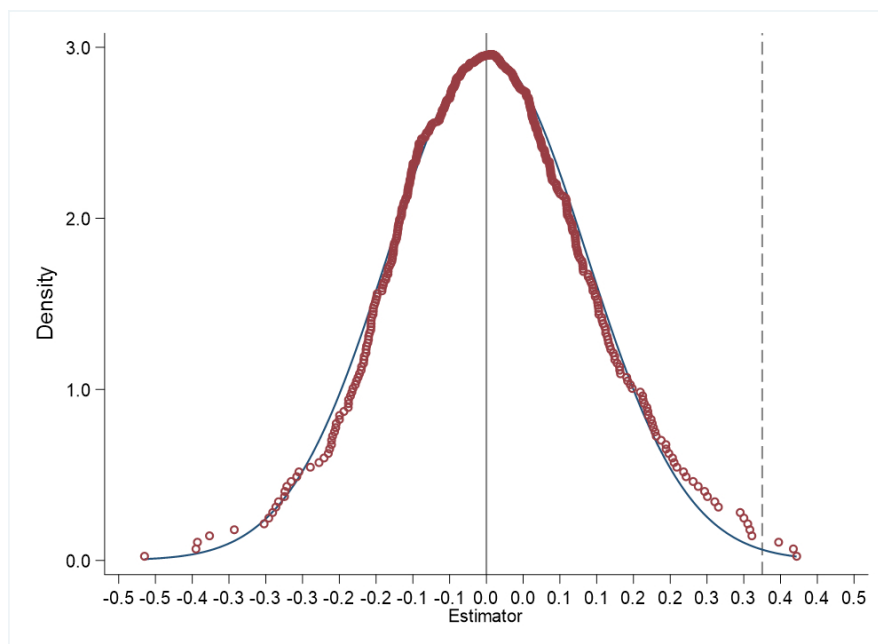


Figure 2: Placebo test

- Sustainability 14, no. 17: 11078. <https://doi.org/10.3390/su141711078>
- [6] Liu, Li, Tao Ding, and Hao Wang. 2022. "Digital Economy, Technological Innovation and Green High-Quality Development of Industry: A Study Case of China" Sustainability 14, no. 17: 11078. <https://doi.org/10.3390/su141711078>
- [7] Nepelski, D. 2019. How to Facilitate Digital Innovation in Europe. *Intereconomics* 54, 47–52. <https://doi.org/10.1007/s10272-019-0791-6>
- [8] Lee G. Branstetter, Raymond Fisman, C. Fritz Foley, Do Stronger Intellectual Property Rights Increase International Technology Transfer? Empirical Evidence from U. S. Firm-Level Panel Data, *The Quarterly Journal of Economics*, Volume 121, Issue 1, February 2006, Pages 321–349, <https://doi.org/10.1093/qje/121.1.321>
- [9] Calignano, G., & Fitjar, R. D. 2017. Strengthening relationships in clusters: How effective is an indirect policy measure carried out in a peripheral technology district?. *The Annals of Regional Science*, 59, 139–169. <https://doi.org/10.1007/s00168-017-0821-x>
- [10] Naghavi, A., & Strozzi, C. 2015. Intellectual property rights, diasporas, and domestic innovation. *Journal of International Economics*, 96(1), 150–161. <https://doi.org/10.1016/j.jinteco.2015.01.007>
- [11] Gangopadhyay, K., & Mondal, D. 2012. Does stronger protection of intellectual property stimulate innovation?. *Economics Letters*, 116(1), 80–82. <https://doi.org/10.1016/j.econlet.2012.01.006>